Chapter 18 Problem Set

Siddharth K

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1. Let A be the set of people taking yoga, B be bridge, and C be painting. By PIE, $|A \cup B \cup C| = |A|+|B|+|C|-|A \cap B|-|A \cap C|-|B \cap C|+|A \cap B \cap C|$. We can rewrite this as $|A|+|B|+|C|-(|A \cap B|+|A \cap C|+|B \cap C|-2|A \cap B \cap C|)-|A \cap B \cap C|$. Substituting for values we know, we get $20 = 10 + 13 + 9 - 9 - |A \cap B \cap C|$, so $|A \cap B \cap C| = 3$.

2. The number of functions with at least k fixed points is $\binom{n}{k} n^{n-k}$, since we need to choose k numbers to be fixed and the rest have n options. By the Principle of Inclusion-Exclusion, we get the answer of $\sum_{k=0}^{n} (-1)^k \binom{n}{k} n^{n-k}$.

Another solution to this is $(n-1)^n$, since every number in the domain can map to any number besides itself, and output values are allowed to repeat. Evaluating $(n-1)^n$ using the Binomial Theorem, we see that it is indeed equal to $\sum_{k=0}^{n} (-1)^k \binom{n}{k} n^{n-k}$.

3. The number of ways to choose a hand of no k jacks, queens, kings, or aces is $\binom{4}{k}\binom{52-4k}{13}$. Using the Principle of Inclusion-Exclusion, we get that the number of ways to choose no jacks, queens, kings, or aces is $\sum_{k=1}^{4} (-1)^{k-1} \binom{4}{k}\binom{52-4k}{13}$. Subtracting that from $\binom{52}{13}$, we get $\sum_{k=0}^{4} (-1)^k \binom{4}{k}\binom{52-4k}{13}$.

4. First, let's consider the integers divisible by 4 and 5 but not 6 or 7. By PIE, this is just $\lfloor 10000/20 \rfloor - \lfloor 10000/60 \rfloor - \lfloor 10000/140 \rfloor + \lfloor 10000/420 \rfloor = 286$.

For the ones divisible by 4 and 6 but not 5 or 7, this is $\lfloor 10000/12 \rfloor - \lfloor 10000/60 \rfloor - \lfloor 10000/84 \rfloor + \lfloor 10000/420 \rfloor = 571$.

For the ones divisible by 4 and 7 but not 5 or 6, we get $\lfloor 10000/28 \rfloor - \lfloor 10000/140 \rfloor - \lfloor 10000/84 \rfloor + \lfloor 10000/420 \rfloor = 190$.

For the ones divisible by 5 and 6 but not 4 or 7, we have $\lfloor 10000/30 \rfloor - \lfloor 10000/60 \rfloor - \lfloor 10000/210 \rfloor + \lfloor 10000/420 \rfloor = 143$.

For the ones divisible by 5 and 7 but not 4 or 6, this is $\lfloor 10000/35 \rfloor - \lfloor 10000/140 \rfloor - \lfloor 10000/210 \rfloor + \lfloor 10000/420 \rfloor = 190$.

Finally, for the ones divisible by 6 and 7 but not 4 or 5, this is $\lfloor 10000/42 \rfloor - \lfloor 10000/84 \rfloor - \lfloor 10000/210 \rfloor + \lfloor 10000/420 \rfloor = 95$.

Adding all these up, we get 1475.

5. Such an excedance set either has $n-1$ or n in the $(n-2)$'th position or n in the $(n-1)$'th position. Let's assume either $n-1$ or n is in the $(n-2)$'th spot. Then, there are $(n-1)!$ ways to order the rest. We get the same answer when we assume n is in the $(n-1)$'th spot. Now, let's assume that both statements are true. Then there are $(n-2)!$ ways to order the rest. Therefore, the answer is $2(n-1)! - (n-2)!$.

6. Let A_n be the set of perfect n'th powers less than 100000. If $n > 16$, $|A_n| = 0$ because $2^{17} > 100000$. Therefore, we only need to consider $2 \leq n \leq 16$. Also, $|A_n| = \lfloor 100000^{\frac{1}{n}} \rfloor - 1$. By the Principle of Inclusion-Exclusion, we get

$$\left| \bigcup_{i=2}^{16} A_i \right| = \sum_{i=2}^{16} |A_i| - \sum_{2 \leq i \leq j \leq 16} |A_i \cap A_j| + \cdots - \left| \bigcap_{i=2}^{16} A_i \right|.$$

We can simplify this to

$$\sum_{i=2}^{16} (\lfloor 100000^{1/i} \rfloor - 1) - \sum_{2 \leq i \leq j \leq 16} (\lfloor 100000^{1/\mathrm{lcm}(i,j)} \rfloor - 1) + \cdots - (\lfloor 100000^{1/\mathrm{lcm}(2,3,\ldots,16)} \rfloor - 1).$$

That looks extremely hard to simplify further.

8. The expression $\sum_{i=0}^{n} \binom{n}{i} d(i)$ could represent the number of ways to choose i elements of $[n]$ and make a derangement of them. Summing over all i from 0 to n gives us all possible permutations of $[n]$, so the sum is equal to $n!$.